

A Comprehensive Measure of Well-Being

Project Description:

The Human Development Index (HDI) Prediction System is a Machine Learning-based web application that predicts the Human Development Index category of a country using important socio-economic indicators. The application analyzes Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI) per Capita to classify a country into one of four HDI categories:

- Very High
- High
- Medium
- Low

The system is developed using Python, Flask, and Scikit-learn. A trained Machine Learning model processes user inputs and predicts the HDI category instantly through a simple web interface. The project aims to support researchers, policymakers, and students in understanding human development across countries.

Scenarios

Scenario 1: A policymaker enters:

- Life Expectancy = 82 years
- Mean Years of Schooling = 13
- Expected Years of Schooling = 17
- GNI per Capita = \$48,000

The system predicts

HDI Category: Very High

Scenario 2: A researcher enters

- Life Expectancy = 72
- Mean Years of Schooling = 10
- Expected Years of Schooling = 13
- GNI = \$11,000

Prediction

HDI Category: High

Scenario 3:

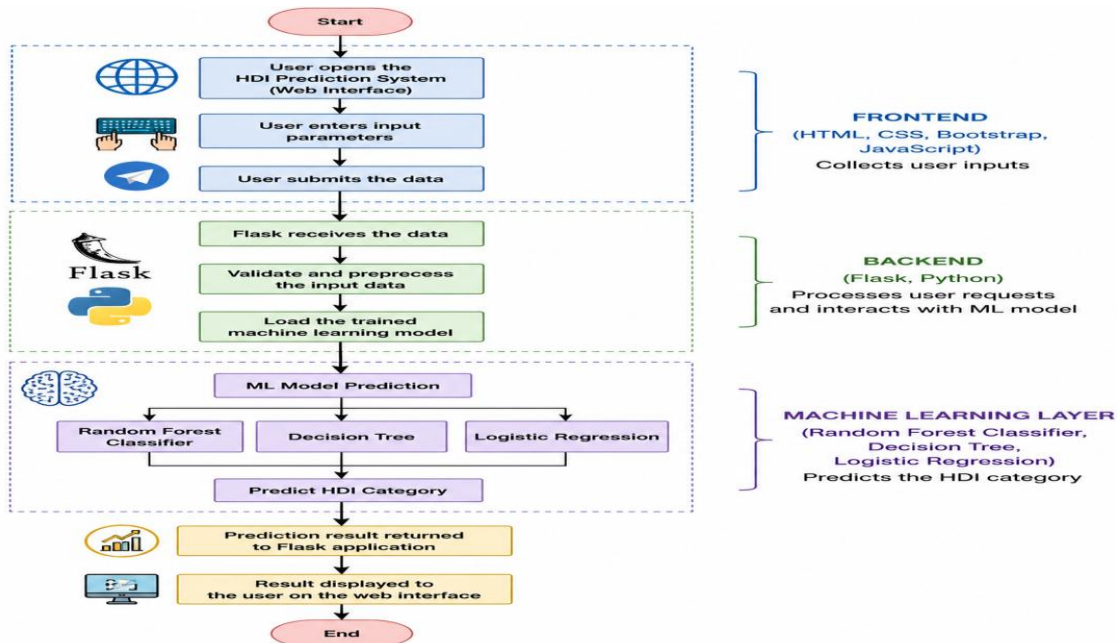
A student enters

- Life Expectancy = 60
- Mean Years of Schooling = 6
- Expected Years of Schooling = 8
- GNI = \$2,500

Prediction

HDI Category: Low

Technical Architecture:



Description:

The prediction result is returned to the Flask application and displayed to the user. This mirrors the architecture section described in the template.

Pre-requisites:

- Python 3.10+
- Flask
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Pickle
- HTML
- CSS
- JavaScript
- VS Code

Project Workflow:

Milestone 1: Problem Identification and Requirement Analysis

Activity 1.1: Identify the Problem

- Study the Human Development Index (HDI) and its importance.
- Analyze the factors affecting HDI prediction.
- Define the project objectives and scope.

Activity 1.2: Requirement Gathering

- Identify required input parameters.
- Collect functional and non-functional requirements.
- Select suitable technologies for development.

Activity 1.3: Technology Selection

- Python
- Flask
- Scikit-learn
- Pandas
- NumPy
- HTML
- CSS
- Bootstrap
- JavaScript

Activity 1.4: Development Environment Setup

- Install Python
- Install VS Code
- Create virtual environment
- Install required libraries using requirements.txt

Milestone 2: Data Collection and Preprocessing

Activity 2.1: Dataset Collection

- Obtain the HDI dataset from a reliable source.

- Analyze dataset structure.

Activity 2.2: Data Cleaning

- Remove missing values.
- Handle duplicate records.
- Check inconsistent data.

Activity 2.3: Data Preprocessing

- Encode categorical variables.
- Normalize/standardize numerical values if required.
- Split dataset into training and testing sets.

Activity 2.4: Exploratory Data Analysis

- Visualize data using graphs.
- Analyze feature relationships.
- Identify important attributes affecting HDI.

Milestone 3: Machine Learning Model Development

Activity 3.1: Model Selection

Compare multiple machine learning algorithms:

- Decision Tree
- Random Forest
- Logistic Regression

Activity 3.2: Model Training

- Train each model using training data.
- Optimize model parameters.

Activity 3.3: Model Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall

- F1 Score
- Confusion Matrix

Activity 3.4: Model Saving

- Select the best-performing model.
- Save the trained model using Pickle.

Milestone 4: Backend Development

Activity 4.1: Flask Application Development

- Create Flask application.
- Configure routes.

Activity 4.2: Load Machine Learning Model

- Load the Pickle model.
- Load preprocessing objects if required.

Activity 4.3: Prediction Logic

- Receive user input.
- Convert input into model format.
- Predict HDI category.
- Return prediction result.

Activity 4.4: Error Handling

- Validate user inputs.
- Handle invalid values.
- Display user-friendly error messages.

Milestone 5: Frontend Development

Activity 5.1: Design User Interface

- Create Home page.
- Design prediction form.
- Build responsive layout using Bootstrap.

Activity 5.2: User Input Form

Collect:

- Country information
- Economic indicators
- Health indicators
- Education indicators
- Other required parameters

Activity 5.3: Result Display

- Display predicted HDI category.
- Show prediction status.
- Present clean output page.

Activity 5.4: Responsive Design

- Desktop compatibility
- Tablet compatibility
- Mobile compatibility

Milestone 6: Model Integration

Activity 6.1: Connect Frontend with Backend

- Submit form data to Flask.
- Receive prediction response.

Activity 6.2: Prediction Processing

- Convert input into DataFrame.
- Send to trained model.
- Generate prediction.

Activity 6.3: Display Results

- Return prediction to webpage.
- Show prediction instantly.

Milestone 7: Testing and Validation

Activity 7.1: Unit Testing

- Test individual modules.

Activity 7.2: Functional Testing

- Verify all project functionalities.

Activity 7.3: Performance Testing

- Measure prediction speed.
- Check response time.

Activity 7.4: User Acceptance Testing

- Validate prediction accuracy.
- Verify usability.

Milestone 8: Deployment

Activity 8.1: Prepare for Deployment

- Create requirements.txt.
- Configure Flask application.

Activity 8.2: GitHub Repository

- Upload source code.
- Maintain project version control.

Activity 8.3: Render Deployment

- Deploy Flask application on Render.
- Verify online accessibility.

Activity 8.4: Live Testing

- Test prediction through deployed website.
- Verify all features work correctly.

Milestone 9: Documentation

Activity 9.1: Prepare Project Report

Include:

- Introduction
- Literature Survey
- System Analysis
- System Design
- Implementation
- Results
- Conclusion

Activity 9.2: Prepare Architecture Diagram

- Frontend
- Flask Backend
- Machine Learning Model
- Prediction Output

Activity 9.3: Prepare Flowchart

- User Input
- Data Validation
- Model Prediction
- Result Display

Milestone 10: Project Completion

Activity 10.1: Final Verification

- Verify all modules.
- Check deployment.

Activity 10.2: Performance Analysis

- Compare model accuracies.
- Select best-performing model.

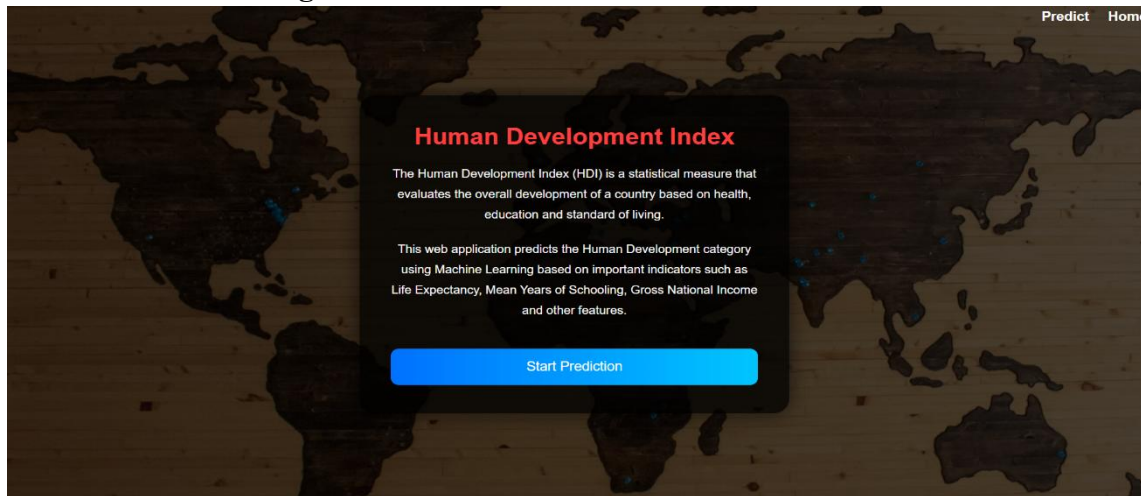
Activity 10.3: Future Enhancements

- Improve prediction accuracy.
- Add more datasets.

- Implement deep learning models.
- Build REST API.
- Add user authentication.
- Store prediction history.
- Deploy on cloud with CI/CD

Exploring the Website's Web Pages:

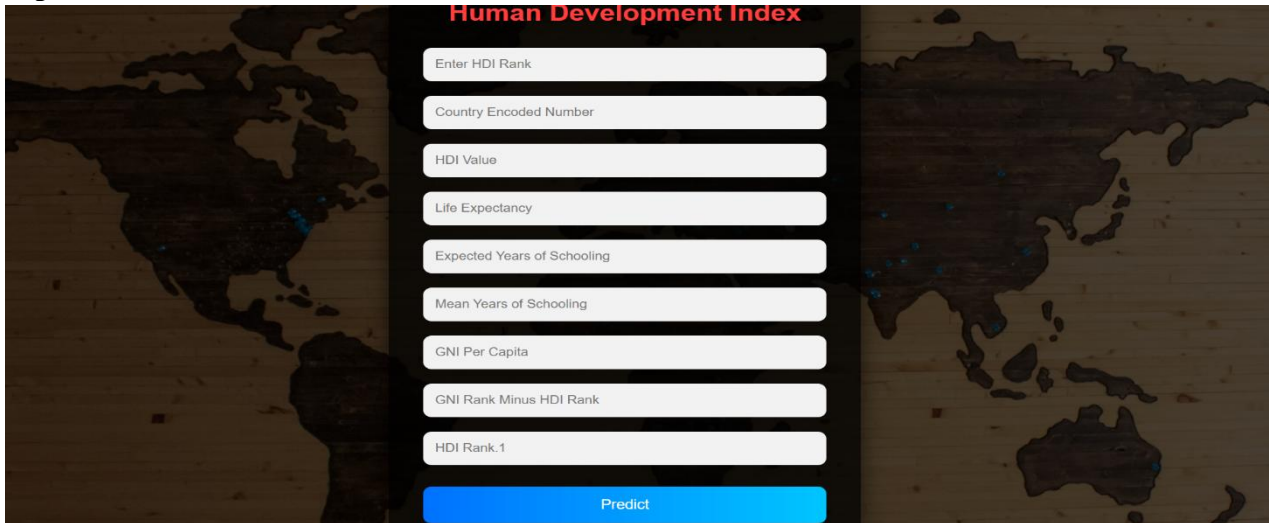
Home / Generator Page:



Description:

The **Home Page** is the landing page of the **Human Development Index (HDI) Prediction System**. It introduces users to the purpose of the application and provides a brief explanation of the Human Development Index. The page explains that HDI is measured using important indicators such as **Life Expectancy, Education, and Gross National Income (GNI)**. It also informs users that the system uses **Machine Learning** to predict the HDI category based on the input values. The page includes a navigation menu and a **Start Prediction** button, allowing users to easily access the prediction page through a simple and user-friendly interface.

Input Section:



The screenshot shows the 'Human Development Index' input section. It features a dark background with a world map. A central black panel contains nine white input fields and a blue 'Predict' button. The input fields are labeled: 'Enter HDI Rank', 'Country Encoded Number', 'HDI Value', 'Life Expectancy', 'Expected Years of Schooling', 'Mean Years of Schooling', 'GNI Per Capita', 'GNI Rank Minus HDI Rank', and 'HDI Rank.1'.

Description: The input section of the HDI Prediction System provides a simple and interactive form where users enter HDI-related parameters such as HDI Rank, Country Encoded Number, HDI Value, Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, GNI per Capita, GNI Rank Minus HDI Rank, and HDI Rank.1 for prediction.

Results Section:



The screenshot shows the 'Human Development Index' results section. It features a dark background with a world map. A central black panel displays the title 'Human Development Index' in red, followed by the subtitle 'A Machine Learning Web Application using Flask' in white. Below this, the prediction result is shown in green: 'Prediction : VERY HIGH HUMAN DEVELOPMENT'. At the bottom of the panel is a blue 'Predict Again' button.

Description: The output section displays the predicted Human Development Index category after processing the entered data using the trained machine learning model. The result is presented

clearly, indicating the predicted classification such as Very High Human Development, along with a Predict Again button that allows users to perform another prediction easily.

Conclusion:

The **Human Development Index (HDI) Prediction System** is an end-to-end Machine Learning application that predicts a country's HDI category using key socio-economic indicators such as life expectancy, education, and income. Built with Python, Scikit-learn, and Flask, it provides fast and accurate predictions through an easy-to-use web interface. The project demonstrates the practical application of machine learning in development analysis and can assist policymakers, researchers, and students in comparing human development across countries and supporting informed decision-making. This mirrors the concluding style of the provided project guide.